

Outline

Linear independence

Basis

Orthonormal vectors

Gram–Schmidt algorithm

Orthonormal vectors

- ▶ set of n -vectors $a_1, \dots, a_k$ are (*mutually*) *orthogonal* if $a_i \perp a_j$ for $i \neq j$
- ▶ they are *normalized* if $\|a_i\| = 1$ for $i = 1, \dots, k$
- ▶ they are *orthonormal* if both hold
- ▶ can be expressed using inner products as

$$a_i^T a_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

- ▶ orthonormal sets of vectors are linearly independent
- ▶ by independence-dimension inequality, must have $k \leq n$
- ▶ when $k = n$, $a_1, \dots, a_n$ are an *orthonormal basis*

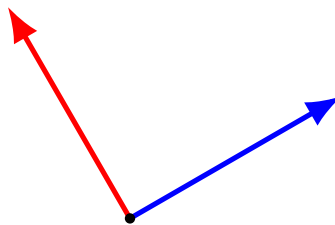
Examples of orthonormal bases

- ▶ standard unit n -vectors $e_1, \dots, e_n$

- ▶ the 3-vectors

$$\begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \quad \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

- ▶ the 2-vectors shown below



Orthonormal expansion

- ▶ if $a_1, \dots, a_n$ is an orthonormal basis, we have for any n -vector x

$$x = (a_1^T x)a_1 + \dots + (a_n^T x)a_n$$

- ▶ called *orthonormal expansion of x* (in the orthonormal basis)
- ▶ to verify formula, take inner product of both sides with a_i

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Gram–Schmidt (orthogonalization) algorithm

- ▶ an algorithm to check if $a_1, \dots, a_k$ are linearly independent
- ▶ we'll see later it has many other uses

Gram–Schmidt algorithm

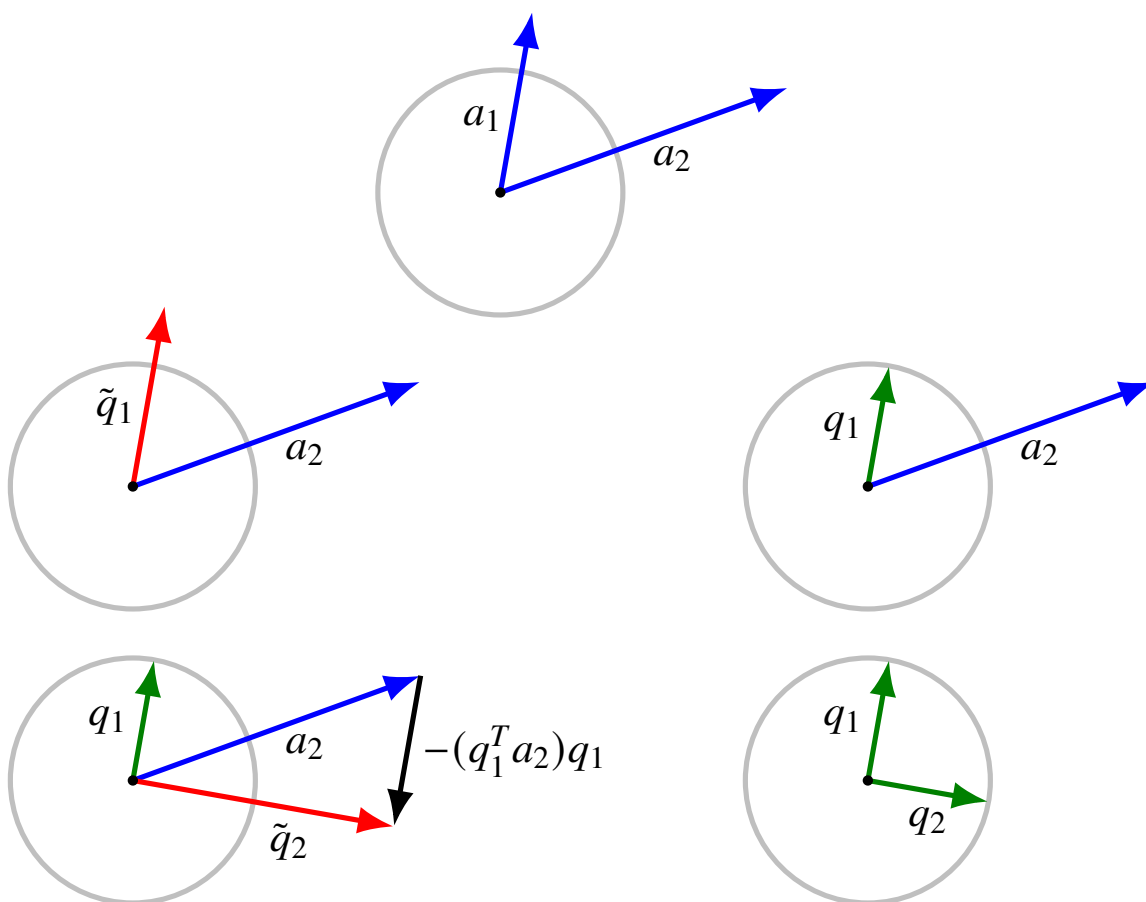
given n -vectors $a_1, \dots, a_k$

for $i = 1, \dots, k$

1. *Orthogonalization*: $\tilde{q}_i = a_i - (q_1^T a_i)q_1 - \dots - (q_{i-1}^T a_i)q_{i-1}$
 2. *Test for linear dependence*: if $\tilde{q}_i = 0$, quit
 3. *Normalization*: $q_i = \tilde{q}_i / \|\tilde{q}_i\|$
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- ▶ if G–S does not stop early (in step 2), $a_1, \dots, a_k$ are linearly independent
- ▶ if G–S stops early in iteration $i = j$, then a_j is a linear combination of $a_1, \dots, a_{j-1}$ (so $a_1, \dots, a_k$ are linearly dependent)

Example



Analysis

let's show by induction that $q_1, \dots, q_i$ are orthonormal

- ▶ assume it's true for $i - 1$
- ▶ orthogonalization step ensures that

$$\tilde{q}_i \perp q_1, \dots, \tilde{q}_i \perp q_{i-1}$$

- ▶ to see this, take inner product of both sides with $q_j, j < i$

$$\begin{aligned} q_j^T \tilde{q}_i &= q_j^T a_i - (q_1^T a_i)(q_j^T q_1) - \dots - (q_{i-1}^T a_i)(q_j^T q_{i-1}) \\ &= q_j^T a_i - q_j^T a_i = 0 \end{aligned}$$

- ▶ so $q_i \perp q_1, \dots, q_i \perp q_{i-1}$
- ▶ normalization step ensures that $\|q_i\| = 1$

Analysis

assuming G–S has not terminated before iteration i

- ▶ a_i is a linear combination of $q_1, \dots, q_i$:

$$a_i = \|\tilde{q}_i\|q_i + (q_1^T a_i)q_1 + \dots + (q_{i-1}^T a_i)q_{i-1}$$

- ▶ q_i is a linear combination of $a_1, \dots, a_i$: by induction on i ,

$$q_i = (1/\|\tilde{q}_i\|) \left(a_i - (q_1^T a_i)q_1 - \dots - (q_{i-1}^T a_i)q_{i-1} \right)$$

and (by induction assumption) each $q_1, \dots, q_{i-1}$ is a linear combination of $a_1, \dots, a_{i-1}$

Early termination

suppose G–S terminates in step j

- ▶ a_j is linear combination of $q_1, \dots, q_{j-1}$

$$a_j = (q_1^T a_j)q_1 + \dots + (q_{j-1}^T a_j)q_{j-1}$$

- ▶ and each of $q_1, \dots, q_{j-1}$ is linear combination of $a_1, \dots, a_{j-1}$
- ▶ so a_j is a linear combination of $a_1, \dots, a_{j-1}$